Problem Summary

Given the root of a binary Search tree, and an integer K, return the Kth Smallest value in the tree.

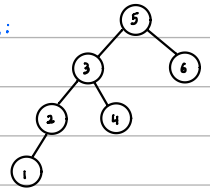
Key Idea

In a BST, inorder traversal visits values in Sorted order: left → root → right

So if we do inorder traversal, the 1st visited node is the smallest, the 2nd visited node is the Second smallest, and so on

Diagram Examples

Before:



Inorder Traversal: 1 → 2 → 3 → 4 → 5 → 6

k=3 ↑ third Smallest

Approach

1) Use inorder traversal: left → root → right

2) Keep track of how many nodes remain before reaching the Kth node

3) When count ==1 : Store the current node value as answer

4) Continue traversal only while count > 0

5) Return the stored answer

Average Worst case

Time Complexity: O(h+k) / O(n) → Inorder traversal may stop after visiting K nodes, but in worst case, we may visit many or all nodes

Space Complexity: O(h) → Recursion stack, where h is the height of the tree → n for skewed tree / → logn for balanced tree